Vishnu Unnikrishnan

Research Data Science @ Pfizer | Master's in Statistics | Cochin University of Science and Technology.
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Experience

Research Intern, Data Science **May 2026 – Nov 2026**
Pfizer India *Chennai, India*

- As Pfizer DSA Data Science Research Intern I worked on developing a novel methodology for high-dimensional feature selection with error control. Adapted the sequential knockoff generation method with a Tweedie distribution model for zero-inflated biomarker data. Built fully reproducible pipelines and ran extensive simulations using R.
- Delivered results on multiple clinical trials, including the **BEACON CRC Trial** and **MagnetisMM Trial**.

Research Intern, Data Science **May 2025 – July 2025**
ICMR - National Institute of Epidemiology *Chennai, India*

- Worked on Multi-state model to stratify the risk of Mucormycosis patients clinical progression in India using semi-Markov proportional hazards model. Built fully reproducible pipeline using `R{mstate, survival}, Excel and Stata.`

Data Science Intern **Dec. 2022 – Jan. 2023**
Kerala State Economics and Statistics Department *Kerala, India*

- Worked on 'Predictive modeling of Paddy Production and Yield' project for the Government of Kerala. Utilized Time Series models to forecast paddy yield based on 4 years of data. Delivered critical data-driven insights for policymaking.

Projects

Multi-Omics Prognostic Biomarker Identification (Master's Thesis) | *R, Machine Learning, Survival Analysis*

- Preprocessed 57K multi-omics features for 439 LUAD patients, deployed Cox-SIS to reduce dimensionality to 288 targets
- Trained penalized Cox & ensemble survival models via 100-subsample Stability Selection to isolate robust biomarkers
- Derived a 9-gene consensus panel & validated risk stratification via cross-validated C-Index & log-rank tests

Stacked Ensemble Prediction of Fatal Myocardial Infarction Complications | *Python, Machine Learning, Data Science*

- Tackled class imbalance using SMOTE & extracted top 22 clinical predictors out of 111 via LASSO & RF-based RFE
- Trained LR, kNN, GBM, RF & XGBoost via 10-fold CV & stacked predictions via GLM & RF meta-learners
- Attained 0.87 Test Accuracy, 0.93 F1 & 0.81 AUC using Stacked RF, with 97.4% sensitivity in identifying fatal cases.

Unsupervised Subgroup Discovery in Leukemia | *Python, Machine Learning*

- Filtered 22k microarray probes to 1k via variance & extracted transcriptomic embeddings using PCA, t-SNE & UMAP
- Clustered PCA space via K-Means, Agglomerative & GMM, attaining 0.85 NMI with BIC-optimized GMM (k=8)
- Identified 988 significant biomarker probes via ANOVA F-testing & Benjamini-Hochberg FDR correction

Survival Analysis of Systolic Blood Pressure & Mortality | *SAS, Survival Analysis*

- Modeled 22-year mortality on 2k Framingham cohort patients via multivariate Cox PH & AIC-driven stepwise selection
- Evaluated PH assumptions via Schoenfeld residuals & time-varying plots, resolving violations via age stratification
- Identified aHR of 1.014 ($p < 0.0001$), indicating a 1.4% mortality risk increase per systolic BP unit

Financial Time Series Forecasting and Risk Analysis | *Python, Time Series Analysis, Deep Learning*

- Engineered 10y TSLA/SPY/BND market data to build ARIMA, SARIMA & LSTM forecasts.

Education

Cochin University of Science and Technology, Kerala, India **July 2024 – July 2026**
Master of Science (M.Sc.), Statistics

St. Thomas College (Autonomous), Kerala, India **Aug. 2020 – July 2023**
Bachelor of Science (B.Sc.), Mathematics

Skills

Technical: Python, R, SAS, Excel, SQL, Power BI, Tableau, Github, Al Agents, Machine Learning, Deep Learning
Interpersonal: Communication, Teamwork, Leadership, Problem-solving
Courses/Certificates: Machine Learning Specialization, R Developer, Statistician in R, Statistics for Clinical Research (SAS, SDTM, ADaM, TFLs) - IQVIA, SAS Programming, Design of Experiments, Data Scientist tech stacks

Achievement / Leadership / Extracurricular

- **State First** (3x) - Maria Philip Memorial Debate Competition (2020,2021,2022). **State Finalist** - Ministry of Youth Affairs and Sports' National Youth Parliament (2023,2024). **Head/President** - Debate and Oratory Club, STC.
- Cleared **GATE ST** (2026), **GATE DA** (2025), IIT JAM MS (2024), **ISI** PGDSMA 2023 (AIR 43)
- Qualified for **National Finals** - **Bitathon Data Science Hackathon** 2026, GIM Goa
- Conference Coordinator & Volunteer Lead - IISA Annual Conference 2024, ICSSA Conference 2026